

Mark Dodici

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RESEARCH INTERESTS

Binary stars in dense clusters; transient phenomena in galactic nuclei; hierarchical triples and few-body dynamics; stellar tides; intermediate-mass black holes; celestial mechanics; galactic dynamics

EDUCATION

University of Toronto

2022 – (expected Aug 2027)

Ph.D. candidate, Department of Astronomy & Astrophysics (UofT DAA)

Graduate researcher, Canadian Institute for Theoretical Astrophysics (CITA)

Visiting Graduate Student, Institute for Advanced Study (Winter 2026)

Advisors: Professors Yanqin Wu & Scott Tremaine

Princeton University

2018 – 2022

A.B. Astrophysical Sciences, *magna cum laude*

Thesis advisors: Dr. Christopher Spalding & Professor Jeremy Goodman

PUBLICATIONS 4 peer-reviewed (2 first-author); 70 citations (33 first-author).

FIRST AUTHOR

5. **M. Dodici**, 2026. Binary disruptions driven by massive disks around massive black holes. [arXiv:2607.26205](https://arxiv.org/abs/2607.26205). Submitted to ApJ. 24 pp.
4. **M. Dodici**, S. Tremaine, & Y. Wu, 2026. Dynamical evolution of stellar binaries in galactic centers. [ApJ, 1000, 226](#). 13 pp.
3. **M. Dodici** & S. Tremaine, 2024. Studying binary formation under dynamical friction using Hill's Problem. [ApJ, 972, 193](#). 26 pp.

CONTRIBUTING

2. B. Hensley, C. Murray, & **M. Dodici**, 2022. Polycyclic Aromatic Hydrocarbons, Anomalous Microwave Emission, and their Connection to the Cold Neutral Medium. [ApJ, 929, 23](#). (9 pp.)

ON EDUCATION

1. B. Lewis et al. (incl. **M. Dodici**), 2025. Improving Undergraduate Astronomy Students' Skills with Research Literature via Accessible Summaries: A Case Study with Astrobites-based Lesson Plans. [Phys. Rev. Physics Education Research, 21, 010124](#). (17 pp.)

TALKS 21 total; 6 invited (*), 1 award-based (†). Upcoming italicized.

AT CONFERENCES

- | | |
|--|------------------------------|
| 21. * Galactic Nuclei Astrophysical Phenomena | Sexten, Italy. Jul 2026 |
| 20. † AAS Division on Dynamical Astronomy 57 | Chicago, IL. Jun 2026 |
| 19. CITA @ 40 / Kingston Meeting 2026 | Toronto, ON. May 2026 |
| 18. Traversing the Galactic Center in Space and Time / IAU Symposium 405 | Brno, Czechia. May 2026 |
| 17. MODEST-25 / IAU Symposium 398 | Seoul, South Korea. Jun 2025 |
| 16. AAS Division on Dynamical Astronomy 56 | Atlanta, GA. May 2025 |
| 15. AAS Division on Dynamical Astronomy 55 | Toronto, ON. May 2024 |
| 14. AAS Division for Planetary Science 55 | San Antonio, TX. Oct 2023 |
| 13. CITA Planet Day | Toronto, ON. Aug 2023 |
| 12. Emerging Researchers in Exoplanet Science Symposium VIII | New Haven, CT. Jun 2023 |

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| 11. Great Lakes Exoplanets Area Meeting | Columbus, OH. Nov 2022 |
| 10. AAS 240 | Pasadena, CA. Jun 2022 |

AT INSTITUTIONS

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| 9. * Tea Talk, Indiana University | Bloomington, IN. Oct 2026 |
| 8. * APS Lunch Seminar, University of Colorado, Boulder | Boulder, CO. Oct 2026 |
| 7. * Astronomy Lunch Seminar, University of California, Los Angeles | Los Angeles, CA. Oct 2026 |
| 6. Galactic Center Group, Max Planck Institute for Extraterrestrial Physics | Garching, Germany. May 2026 |
| 5. Rossi Group, Leiden Observatory | Leiden, Netherlands. May 2026 |
| 4. * Dynamics Blackboard Talk, Institute for Advanced Study | Princeton, NJ. Mar 2026 |
| 3. * Astrophysics Seminar, Institute for Advanced Study | Princeton, NJ. Feb 2026 |
| 2. Compact Objects Group, CITA | Toronto, ON. Feb 2024 |
| 1. Lunch Talk, UofT DAA | Toronto, ON. Jan 2024, Apr 2026 |

FELLOWSHIPS, GRANTS, and AWARDS

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| Jui Lin (Allen) Yen Award (UofT DAA, for outstanding student research) | 2026 |
| Lachlan Gilchrist Fellowship (UofT School of Graduate Studies) | 2026 |
| Duncombe Student Research Prize (AAS Division on Dynamical Astronomy) | 2026 |
| Carl Reinhardt Fellowship in Astronomy (UofT DAA) | 2025 |
| Dunlap Institute Seed Funding Grants for outreach workshop series | 2023, 2024 |
| Graduate Scholarship (CITA) | 2023 |

LEADERSHIP and SERVICE ROLES

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| Referee, <i>ApJ</i> , <i>ApJL</i> , <i>OJA</i> | 2025 – |
| Coding the Cosmos (HS workshop series), <i>Co-director</i> (2024 –) | 2023 – |
| UofT Graduate Astronomy Students Association | 2022 – |
| <i>Co-President</i> (2024 – 2025) | |
| <i>Courses & Qualifying Exams Committee</i> (2022 – 2024, 2025 –) | |
| Department Jamboree, <i>Organizer & Emcee</i> | 2023 – 2025 |
| UofT AstroTours (Public lecture/telescope events), <i>Co-director</i> (2023 – 2024) | 2022 – 2024 |
| Age of the Universe (HS workshop), <i>Co-organizer & Speaker</i> | 2023 |

TEACHING ASSISTANT EXPERIENCE

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| Stars and Planets (AST221, Toronto) | Fall 2024, 2025, 2026 |
| The Sun and its Neighbours (AST101, Toronto) | Fall 2022, 2023 |
| <i>Lecturer, Summer 2026</i> | |
| Introduction to Astrophysics (AST320, Toronto) | Winter 2025 |
| Stars and Galaxies (AST201, Toronto) | Winter 2023, 2024 |
| The Universe (AST203, Princeton) | Winter 2021, 2022 |

OTHER OUTREACH

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| UofT AstroTours, <i>Speaker</i> | 2026 |
| Pursue STEM (HS science program), <i>Workshop leader</i> | 2025 |
| AstroBites (Research blog), <i>Author & Education Study co-author</i> | 2023 – 2025 |
| Eclipse 2024 at the Toronto Public Library (Public lecture series), <i>Speaker</i> | 2024 |
| Solar Eclipse 2024: Indigenous Knowledge (Video), <i>Editor</i> | 2024 |
| Cosmos from your Couch (Astronomy video series), <i>Editor</i> | 2023 |
| ComSciCon Canada (Science Communication Workshop), <i>Attendee</i> | 2023 |
| Princeton Research Day (Virtual talk competition), <i>Award-winning presenter</i> | 2021 |